

Personal Contribution (Evgenii Marenin)

My primary contribution was designing and executing the QGIS-based pipeline. I focused on developing an Object-Based Image Analysis (OBIA) model. My tasks were the following:

- **Data preparation:** I researched the data sources, utilized QuickOSM plugin to establish the regions containing the by generating dissolved buffer corridors around the mapped street network.
- **Feature Engineering & Masking:** I computed the NDVI to mask vegetation and concatenated the original 4-band DOP10 imagery (RGB + NIR) with the NDVI layer into a 5-band raster, which was critical for enhancing the spectral separability between painted lanes, asphalt, and bare soil.
- **Classification:** I manually digitized training sample polygons across target classes, configured and trained a Random Forest classifier using the dzetsaka plugin.
- **Segmentation and vectorization:** applying the MeanShift segmentation algorithm (via Orfeo ToolBox) to the raw imagery, then using Zonal Statistics (Majority rule) to assign the dominant Random Forest class to each structurally homogeneous vector polygon, followed by morphological smoothing.
- **Automation:** I have tried to translate the entire step-by-step pipeline into a single algorithm using the QGIS Graphical Modeler, but unfortunately the result was not successful.

Main Challenges and Solutions

1. **Handling NoData array crashes:** I diagnosed that the Scikit-Learn backend of dzetsaka could not process the NoData voids created by my initial shadow and OSM masks. Instead, I ran the classification on the full, unclipped raster.
2. **Modeler compatibility:** While automating the workflow, the Graphical Modeler continuously failed with "Model not found" errors. Apparently, dzetsaka could not parse QGIS's default temporary file extensions (.file).